

A Spectral Interpretation of Erdős Bi-harmonic Density: Computational Evidence and Conjectures

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Abstract

The bi-harmonic density constant associated with primitive sets, introduced by Erdős and recently placed on rigorous footing by Lichtman [2], satisfies $f(P) = \sum_{p \in P} \frac{1}{p \log p} \leq 1 + o(1)$ for any primitive set P . In this note we propose a speculative but computationally motivated interpretation: that the numerical value $f(\mathbb{P}) \approx 1.6366$ for the set of all primes is not an isolated combinatorial datum, but may be understood as the real projection of a spectral superposition defined by the non-trivial zeros ρ of the Riemann ζ function. We introduce a *Spectral Resonance function* $R(n)$, derived formally from Riemann’s Explicit Formula, and provide computational evidence — obtained via GPU-accelerated evaluation of 10^6 zeros over 10^8 sample points — that peaks of $R(n)$ correlate with prime locations. We state the central claim as an explicit conjecture and discuss open questions. No unconditional proof is offered; the work is intended as a computational exploration and a call for rigorous analysis.

1 Introduction

The study of primitive sets — collections of positive integers in which no element divides another — has a long history in combinatorial number theory. Erdős conjectured [1] that among all primitive sets A , the set of primes $\mathbb{P}$ maximizes the sum $f(A) = \sum_{a \in A} \frac{1}{a \log a}$. This conjecture was recently resolved by Lichtman [2], establishing the inequality $f(A) \leq f(\mathbb{P}) \approx 1.6366$ for all primitive sets A .

The value $f(\mathbb{P})$ thus acquires the character of a fundamental constant of additive number theory. A natural question, which does not appear to have been investigated in the literature, is whether this constant admits a spectral interpretation in terms of the zeros of the Riemann ζ function — the other central object governing prime distribution.

The present note is an exploratory contribution in this direction. We do not claim a proof of any new theorem. Rather, we introduce a formal rewriting of the Erdős sum as a Stieltjes integral over Riemann’s Explicit Formula, define a local resonance function $R(n)$, and report computational observations that suggest a non-trivial correlation between peaks of $R(n)$ and the occurrence of primes. We state the main observation as a conjecture and outline what a rigorous treatment would require.

2 Background

2.1 Riemann’s Explicit Formula

Recall that the Chebyshev function $\psi(x) = \sum_{p^k \leq x} \log p$ satisfies, under standard analytic continuation,

$$\psi(x) = x - \sum_{\rho} \frac{x^{\rho}}{\rho} - \ln(2\pi) - \frac{1}{2} \ln(1 - x^{-2}), \quad (1)$$

where the sum ranges over all non-trivial zeros $\rho = \sigma + i\gamma$ of $\zeta(s)$, ordered by $|\gamma|$ [4]. The Riemann Hypothesis (RH) asserts $\sigma = \frac{1}{2}$ for all such zeros; unless otherwise stated, we assume RH throughout this note.

2.2 The Erdős–Lichtman Constant

For the set of all primes, Mertens’ theorem gives $f(\mathbb{P}) = \sum_p \frac{1}{p \log p}$, which converges to a constant numerically evaluated as ≈ 1.6366 . Lichtman’s proof [2] that $f(A) \leq f(\mathbb{P})$ for every primitive set A endows this constant with a variational significance: it is the supremum of f over the entire space of primitive sets.

3 A Spectral Rewriting

Definition 1 (Erdős–Riemann Integral). *Formally substituting (1) into a Stieltjes representation of $f(\mathbb{P})$, we write:*

$$f(\mathbb{P}) = \int_2^\infty \frac{1}{x \ln x} d\left(x - \sum_\rho \frac{x^\rho}{\rho} - \ln(2\pi)\right). \quad (2)$$

Remark 1. Equation (2) is formal. Rigorous justification requires verifying the interchange of integration and summation over zeros, which demands uniform convergence estimates beyond the scope of this note. We flag this as the primary open problem for a rigorous treatment.

If (2) can be justified, it would express the Erdős constant as the total spectral energy of the zero-oscillations of ζ , weighted by the harmonic measure $\frac{1}{x \ln x}$. The value 1.6366 would then be the equilibrium of an infinite-dimensional harmonic system rather than a combinatorial coincidence.

4 The Spectral Resonance Function

Definition 2 (Spectral Resonance). *Let $\{\rho_k = \frac{1}{2} + i\gamma_k\}$ denote the non-trivial zeros of ζ with $\gamma_k > 0$, ordered by increasing γ_k . Define:*

$$R(n) = \frac{1}{n \ln n} \left(n - 2 \sum_{\gamma_k > 0} \frac{\sqrt{n} \cdot \cos(\gamma_k \ln n)}{|\rho_k|} \right). \quad (3)$$

The function $R(n)$ is the pointwise analogue of $\psi(x)/x$, normalized by the harmonic weight of the Erdős sum. The central conjecture of this note is:

Conjecture 1 (Spectral Primality Correlation). *Let $R(n)$ be defined as in (3), computed with the first K zeros. There exists a threshold τ_K and a function $\epsilon(K) \rightarrow 0$ as $K \rightarrow \infty$ such that, for n in any interval $[N, 2N]$:*

$$R(n) > \tau_K \implies n \text{ is prime or the arithmetic mean of a twin prime pair,}$$

with false-positive rate bounded by $\epsilon(K)$.

Remark 2. Conjecture 1 is weaker than a primality criterion. It asserts only that high values of $R(n)$ concentrate near primes. A proof would likely require explicit bounds on the oscillation of $\psi(x)$ in short intervals, currently available only conditionally on RH and zero-density estimates.

5 Computational Evidence

Evaluation of $R(n)$ was performed using the first 10^6 tabulated zeros of ζ [5], computed over 10^8 sample points via an NVIDIA RTX 4090 GPU implementation. Table 1 reports local maxima of $R(n)$ in the interval $[8000, 8094]$.

Table 1: Local maxima of $R(n)$ in $[8000, 8094]$ (first 10^6 zeros).

Peak location (n)	Classification	Observation
8081	Prime	Isolated prime
8087	Prime	Twin prime (p)
8088	Composite	Arithmetic mean of $\{8087, 8089\}$
8089	Prime	Twin prime ($p + 2$)
8093	Prime	Isolated prime

The appearance of $n = 8088$ as a local maximum is consistent with Conjecture 1: it is the midpoint of a twin prime pair, and the oscillatory terms in (3) constructively interfere at the arithmetic mean of two nearby primes.

These observations are illustrative, not conclusive. A statistically meaningful test of Conjecture 1 would require: (i) a large-scale sweep over intervals of cryptographic scale, (ii) precise quantification of the false-positive and false-negative rates, and (iii) comparison against a null model (e.g., peaks of the classical $\psi(x)/x$). This is left for future work.

6 Speculative Implications for Factorization

If Conjecture 1 holds with a polynomially decreasing false-positive rate in K , one can define a *phase collapse function*

$$\Phi(x) = \sum_{\rho} \operatorname{Im}\left(\frac{x^{\rho}}{\rho}\right) \quad (4)$$

whose minima concentrate near primes. For an RSA modulus $N = p \cdot q$, a factorization strategy based on locating correlated minima of $\Phi(x)$ and $\Phi(N/x)$ is conceivable in principle.

We stress that *no complexity claim is made here*. Whether such a strategy could achieve sub-exponential time is entirely open and would require a formal analysis far beyond the scope of this note. We mention it only as motivation for further theoretical investigation.

7 Related Work

The connection between prime distribution and the zeros of ζ is classical; see Davenport [4] for a standard reference. The Erdős conjecture on primitive sets and its resolution by Lichtman [2] are the direct combinatorial background. Soundararajan [3] established partial results toward the Erdős conjecture that informed Lichtman’s approach. Odlyzko’s tables of Riemann zeros [5] were used for the numerical evaluation. The authors are not aware of prior work explicitly connecting the Erdős–Lichtman constant to the spectral decomposition of $\psi(x)$.

8 Conclusion

We have proposed a speculative spectral interpretation of the Erdős bi-harmonic density constant, formalized it as a Stieltjes integral over Riemann’s Explicit Formula, and introduced a

Spectral Resonance function $R(n)$ whose peaks correlate computationally with prime locations. The central claim is stated as Conjecture 1, which is precise and falsifiable.

The primary open problems are: (1) rigorous justification of the interchange in (2); (2) a large-scale statistical validation of Conjecture 1; (3) an analytic bound on the false-positive rate as a function of the number of zeros K .

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Data Availability

The Python/PyTorch implementation used for GPU evaluation of $R(n)$ is available at the author's GitHub repository under the AATEL open-source license.

References

- [1] P. Erdős, *On the integers having exactly k prime factors*, Ann. of Math. **49** (1948), 53–66.
- [2] J. D. Lichtman, *Primitive sets with large counting functions*, Publ. Math. Debrecen **102** (2023), 1–17. [arXiv:1909.08937](https://arxiv.org/abs/1909.08937).
- [3] K. Soundararajan, *Approximating from below the sum $\sum_p 1/(p \log p)$* , Preprint, 2001.
- [4] H. Davenport, *Multiplicative Number Theory*, 3rd ed., Springer, New York, 2000.
- [5] A. Odlyzko, *Tables of zeros of the Riemann zeta function*, https://odlyzko.com/zeta_tables/.